

Q. Let $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 2 \end{bmatrix}$ then compute M_{12} , M_{22} , M_{31} , A_{22} and A_{31}

sol. First find Minors.

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 2 \end{bmatrix}$$

Now find M_{12} Means first row skip and select second column.

$$\text{so } M_{12} = \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix}$$

then find determinant of it.

$$M_{12} = (2)(1) - (1)(1)$$

$$M_{12} = 2 - 1$$

$$\boxed{M_{12} = 1} \rightarrow \text{determinant.}$$

$$\text{Now } M_{22} \quad A = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 2 \end{vmatrix}$$

second row second column.

$$M_{22} = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix}$$

$$4 - 1$$

$$\boxed{M_{22} = 3}$$

same for

$$M_{21} = \begin{vmatrix} 1 & 1 \\ -1 & 2 \end{vmatrix}$$

$$M_{21} = 3$$

Now find co-factors.

columns. $i+j \rightarrow$ we will add rows and columns.
 $A_{ij} = (-1)^{i+j}$ then M_{ij}
rows
Any Matrix

if A_{21} then $= (-1)^{2+1}$ and M_{21} which is minor
same minor we will multiply it here.

Means if co-factor is A_{22} then minor M_{22}
must be exists because we will multiply it
with minor
sol

Now we have to find co-factor of
 $A_{12} = (-1)^{1+2} M_{12}$

$$A_{12} = (-1)^3 (1)$$

$$A_{12} = -1 \times 1$$

$$A_{12} = -1$$

same for A_{21}

$$A_{21} = (-1)^{2+1} (3)$$

$$A_{21} = -3$$

Now same for A_{22}

$$A_{22} = (-1)^{2+2} M_{22}$$

$$A_{22} = (1) (3)$$

$$A_{22} = 3$$